SIMATS ENGINEERING

ASSESSMENT TOOL 2 : Scenario-Based

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Assessment Tool Weightage: 20%

Dave's Taxonomy: Manipulation (Level 2) **Total Marks: 25**

Question Count: 5

Code of Conduct

I **Nevitha Magendran/192521206** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: **Nevitha**

Assessment Tasks

Scenario-Based Questions

1. A media company wants to migrate its video streaming platform to the cloud. The system must handle millions of concurrent users with minimal buffering. Design a cloud architecture using scalability and caching techniques. Explain how load balancing and auto-scaling can ensure smooth streaming. Discuss cost optimization strategies for such high-demand workloads.

   Answer:

   Proposed architecture: Users connect through a global CDN (e.g., CloudFront/Akamai) whose edge servers cache popular video segments. Requests that miss the cache go to a Global Load Balancer, then to an Application Load Balancer in the nearest region. Behind it, stateless web/API servers (in an auto-scaling group or Kubernetes) handle login, search and recommendations. Raw and encoded videos are stored in object storage (S3/Blob), a media-processing pipeline transcodes each video into multiple resolutions (adaptive bitrate HLS/DASH), metadata sits in a replicated NoSQL/SQL database with an in-memory cache (Redis), and monitoring tools watch the whole stack.

Scalability and caching techniques: Horizontal scaling of stateless servers, multi-layer caching (browser, CDN edge, regional cache, Redis for metadata), adaptive bitrate streaming so that video quality adjusts to the viewer's bandwidth, and pre-positioning (pre-fetching) of new releases to edge nodes before launch.

Load balancing and auto-scaling for smooth streaming: The load balancer spreads millions of concurrent requests across healthy servers, using health checks and geo-routing to send each viewer to the closest region. Auto-scaling adds instances when CPU, bandwidth or concurrent-session metrics cross a threshold (and uses scheduled scaling before big events such as a live match or series release) and removes them afterwards. Together they prevent overload, so viewers see minimal buffering and no outages.

Cost optimization: Serve most traffic from CDN caches to reduce origin and data-transfer costs; use reserved/committed instances for the steady base load, on-demand for spikes, and spot instances for transcoding jobs; move older or rarely watched videos to cheaper storage tiers using lifecycle policies; right-size instances; compress and use efficient codecs (H.265/AV1); set budgets and use cost-monitoring tools to track usage.

2. A fintech startup is deploying applications using containers across multiple cloud providers. They face challenges in maintaining consistency and managing deployments. Propose a solution using container orchestration and multi-cloud strategies. Explain how service discovery and scaling can be handled efficiently. Highlight the benefits and risks of a multi-cloud deployment model.

   Answer:

   Proposed solution: Package every service as a container image (Docker) and store it in a common registry. Use Kubernetes as the standard orchestration layer on every provider (EKS on AWS, AKS on Azure, GKE on Google Cloud), optionally managed centrally through a multi-cluster tool such as Rancher, Anthos or Azure Arc. Describe the infrastructure with Infrastructure-as-Code (Terraform) and the applications with Helm charts / YAML manifests, and deploy through a single CI/CD pipeline using GitOps (Argo CD / Flux). The same manifests and images are then applied identically in every cloud, which solves the consistency problem and gives repeatable, automated deployments.

   Service discovery and scaling: Kubernetes gives each service a stable DNS name and virtual IP (Service objects), so containers find each other even when pods are restarted or moved; a service mesh (Istio/Linkerd) or Consul can extend discovery and secure (mTLS) traffic across clusters and clouds. For scaling, the Horizontal Pod Autoscaler adds or removes pods based on CPU or custom metrics, the Cluster Autoscaler adds or removes nodes, and global DNS / traffic managers route users to the best cluster.

Benefits of multi-cloud: avoids vendor lock-in, improves resilience (an outage at one provider does not stop the business), allows the best service or price from each provider, helps meet regional/regulatory requirements, and gives better negotiating power.

Risks: higher operational complexity and skill requirements, inconsistent security and IAM models across providers, data-transfer (egress) charges and latency between clouds, difficulty in unified monitoring and cost tracking, and a larger attack surface. These are reduced by using standard tools (Kubernetes, Terraform), centralised logging/monitoring, and a common security policy.

3. An e-learning platform experiences data loss due to accidental deletion in the cloud. The organization needs a robust backup and disaster recovery strategy. Design a solution using cloud-native backup, replication, and versioning. Explain how recovery time objective (RTO) and recovery point objective (RPO) are achieved. Provide steps to ensure business continuity during failures.

   Answer:

   Solution design: Course videos, documents and student submissions are kept in object storage with versioning enabled and MFA-delete / object-lock, so an accidental deletion only adds a delete marker and the earlier version can be restored. Databases use automated snapshots plus point-in-time recovery (PITR), and block-storage volumes are snapshotted on a schedule using a backup service (AWS Backup / Azure Backup). Data is replicated to a second Availability Zone and, through cross-region replication, to another region. Backups are encrypted, protected by least-privilege IAM, and managed with retention and lifecycle policies. Restores are tested regularly.

   Achieving RTO and RPO: RPO (how much data can be lost) is met by the frequency of backups and replication — continuous replication and PITR give an RPO of seconds to minutes, hourly or daily snapshots give an RPO of hours. RTO (how quickly service must be restored) is met by the recovery design — a warm standby or pilot-light environment in the second region, infrastructure defined as code, automated restore scripts and DNS failover bring the platform back in minutes, whereas restoring from backup only takes longer. For example, a target of RPO = 15 minutes and RTO = 1 hour needs PITR/replication plus a warm standby.

   Steps for business continuity: (1) identify critical systems and set RTO/RPO targets; (2) enable versioning, backups and replication; (3) write and document a disaster recovery plan with roles and contacts; (4) monitor and alert on failures or unusual deletions; (5) on failure, declare the incident, fail over to the standby region or restore the last good version; (6) verify data integrity and redirect users through DNS; (7) communicate status to students and staff; (8) after recovery, run a review, fix the root cause and test the plan again with periodic DR drills.

4. A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data. Some applications must remain on-premises due to compliance requirements. Design a hybrid architecture that ensures secure communication between environments. Explain how identity management and encryption should be implemented. Discuss challenges in managing hybrid cloud environments.

   Answer:

   Hybrid architecture: Sensitive and compliance-bound applications and data stay in the on-premises data centre (private cloud), while non-sensitive workloads such as public portals, analytics and development run in the public cloud. The two are linked by a dedicated private connection (AWS Direct Connect / Azure ExpressRoute) with a site-to-site IPsec VPN as backup. Traffic passes through firewalls, a DMZ and segmented networks (VPC/VNet with private subnets), an API gateway controls which services can be called, and only the minimum required data is shared with the cloud (data classification decides what may move).

   Identity management: Use one central identity provider (e.g., Active Directory federated with Azure AD / cloud IAM) for single sign-on, enforce multi-factor authentication, role-based access control with least privilege, privileged-access management, and regular access reviews. Secrets and service accounts are held in a vault.

   Encryption: Encrypt data in transit with TLS and the IPsec/MACsec link, and at rest with AES-256. Keep encryption keys under the agency's control in an on-premises HSM or customer-managed KMS keys (BYOK/HYOK), rotate them regularly, and tokenise or mask highly sensitive fields.

   Challenges: complex integration of different platforms and APIs; maintaining consistent security policies, compliance and auditing in both environments; network latency and bandwidth between sites; unified monitoring, logging and visibility; data-movement and synchronisation issues; skill gaps and higher operational cost; and avoiding shadow IT and configuration drift.

5. A SaaS provider needs to ensure high availability and fault tolerance for its application. The system must continue operating even if one region goes down. Propose a multi-region deployment strategy in the cloud. Explain how data replication and failover mechanisms should be configured. Discuss monitoring and alerting strategies for proactive issue detection.

   Answer:

   Multi-region strategy: Deploy the full application stack in at least two regions (e.g., Region A and Region B) using Infrastructure-as-Code so both are identical. Each region has its own load balancer, auto-scaling application tier spread over several Availability Zones, and database. A global traffic manager (Route 53 / Azure Front Door / Cloud Load Balancing) with health checks routes users to the

nearest healthy region. The model can be active-active (both regions serve users, best availability and performance) or active-passive (secondary on warm standby, lower cost).

Data replication and failover configuration: Use cross-region replication for the data tier — managed databases with read replicas or global tables (e.g., Aurora Global Database, DynamoDB global tables, Cosmos DB multi-region), and replicated object storage. Asynchronous replication gives low latency with a small RPO, synchronous replication gives zero data loss but higher latency. For failover, configure health checks on the endpoints, short DNS TTLs and automatic failover policies so that when the primary region fails traffic shifts to the secondary region and the replica is promoted to primary. Keep configuration, secrets and container images replicated, and test failover regularly with game-day drills.

Monitoring and alerting: Collect metrics, logs and traces (CloudWatch/Azure Monitor/Prometheus, ELK, Datadog) on latency, error rate, CPU, replication lag and database health. Use synthetic checks from several locations to detect outages before customers do, set threshold and anomaly-based alarms, send alerts to the on-call team via email/SMS/PagerDuty with defined escalation, display real-time dashboards, and link alarms to automated remediation (auto-scaling, restart, failover). Define SLOs and review incidents to improve.

Scenario-Based Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Scenario	20%	Demonstrates complete understanding of the scenario context	Good understanding with minor gaps	Partial understanding	Fails to understand the scenario
Identification of Risks / Issues	20%	Identifies all major issues correctly	Identifies most issues	Limited issue identification	Misses major issues
Application of Concepts	25%	Applies virtualization concepts effectively	Applies concepts with minor mistakes	Partially correct application	Weak application
Decision Justification	20%	Decisions are well justified and technically strong	Reasonable justification	Weak justification	No useful justification
Clarity	15%	Clear and well-organized response	Generally clear	Somewhat unclear	Poorly presented